

Motion-Guided Multi-Scale Fusion for High-FPS Video Frame Interpolation

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Abstract

Video frame interpolation (VFI) aims to synthesize intermediate frames between consecutive input frames. In large-motion video frame interpolation, the input frames can be sampled from high-frame-rate videos with enlarged temporal gaps, making the task particularly challenging because of large motion displacements and complex occlusions. We observe that different spatial regions within a single video frame may benefit from interpolation at different scales: a uniform input scale can lead to inaccurate motion estimation in large-motion regions or over-smoothed textures in near-static regions. Motivated by this observation, we propose a motion-guided multi-scale fusion framework that performs frame interpolation at multiple input resolutions using EMA-VFI as the backbone, and adaptively fuses the scale-specific predictions via a learned fusion module conditioned on optical-flow-derived motion maps. A lightweight U-Net predicts spatially varying per-scale fusion weights together with a residual correction, enabling the model to emphasize the most reliable scale in each region according to local motion complexity. The proposed method achieves favorable distortion-perception trade-offs under large-motion settings. Our solution ranked the 1st in both the Development and Test phases of the NTIRE 2026 High-FPS Video Frame Interpolation Challenge – Track 1 (Classic).

1. Introduction

Video frame interpolation (VFI) aims to synthesize temporally intermediate frames given two or more consecutive input frames, and serves as a core component in slow-motion video generation and frame-rate up-conversion [6, 15, 16]. Modern deep-learning-based VFI methods have achieved

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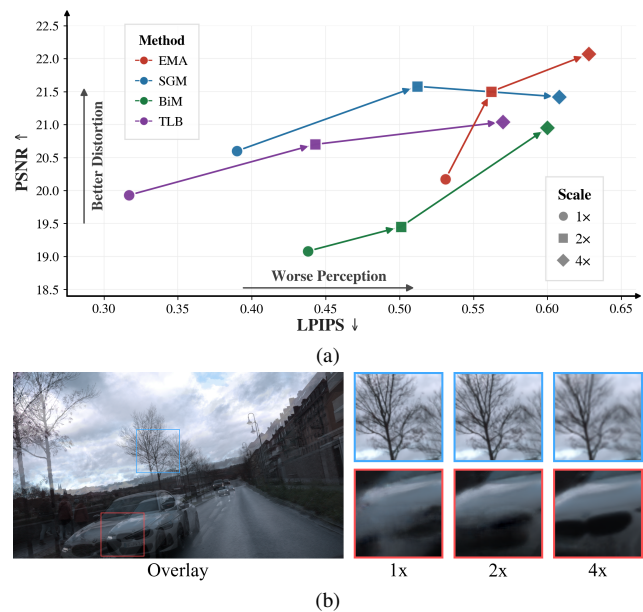


Figure 1. Motivation for motion-guided multi-scale fusion. (a) PSNR vs. LPIPS [27] trade-off for EMA-VFI [26], XvFI [22], SGM [11], BiM-VFI [20], and TLB-VFI [14] at three input scales (1×, 2×, 4× downsampling) on a large-motion subset from the NTIRE 2026 challenge training data. No single scale dominates both fidelity and perceptual quality. (b) EMA-VFI visual example. Larger downsampling ratios better restore large-displacement structures, but blur fine textures after upsampling.

impressive quality on benchmarks such as Vimeo90K [25] and SNU-FILM [1]. Following the NTIRE challenge terminology, we use High-FPS VFI to denote interpolation on frame pairs sampled from high-frame-rate videos. In this setting, the main difficulty comes from enlarged temporal gaps created by frame skipping, which lead to large motion displacements and frequent occlusions.

With respect to the motion magnitude they target, existing VFI methods can be broadly grouped into two categories. General-purpose VFI methods, including flow-

based approaches such as RIFE [4] and EMA-VFI [26], as well as kernel-based and synthesis-based approaches [9, 15], are often evaluated on benchmarks such as Vimeo90K and SNU-FILM. Methods designed for large motion, such as FILM [18] and SGM [11], introduce dedicated modules (e.g., multi-scale feature pyramids or global matching) to handle displacements that exceed the receptive field of common VFI architectures. Among general-purpose methods, EMA-VFI [26] combines CNN and Transformer components to jointly extract motion and appearance features via inter-frame attention, and is a representative recent hybrid CNN-Transformer VFI model.

In the context of extreme-motion interpolation, we start from a straightforward intuition: *downsampling is an effective way to enlarge the receptive field of a conventional VFI model*, making large-displacement content easier to handle. We verify this intuition by running several representative methods—EMA-VFI [26], XVFI [22], SGM [11], BiM-VFI [20], and TLB-VFI [14]—at different input scales ($1\times$, $2\times$, $4\times$ downsampling) and measuring both PSNR and LPIPS on a large-motion subset of the challenge training data. As shown in Fig. 1(a), the trade-off scatter reveals that downsampling improves fidelity in high-motion regions by enlarging the effective receptive field, yet sacrifices perceptual quality in low-motion regions due to the loss of fine-grained texture information. Figure 1(b) further shows that coarser scales can better restore large-displacement structures, but may blur local details after re-upsampling. Crucially, no single scale dominates across both metrics and all spatial regions.

This observation motivates our core idea: rather than committing to a single input scale, we perform frame interpolation at *multiple scales* and *adaptively fuse* the resulting predictions using motion cues derived from optical flow. Specifically, we build a motion-guided multi-scale fusion framework on top of EMA-VFI, in which the input frames are processed at three scales ($1.0\times$, $0.5\times$, $0.25\times$). Each scale branch produces an interpolated frame and bidirectional optical flow at the original resolution. A lightweight U-Net [19] then takes the concatenated multi-scale predictions together with flow-derived motion maps as input, and predicts per-pixel fusion weights and a residual image. The final output is obtained by weighted fusion followed by residual correction, allowing the model to emphasize the most reliable scale in each spatial region according to local motion complexity.

In summary, we propose a motion-guided multi-scale fusion method for High-FPS video frame interpolation. The method is designed to address the scale-dependent distortion-perception trade-off that arises in large-motion settings. Our solution ranked the 1st in both the Development and Test phases of the NTIRE 2026 High-FPS Video Frame Interpolation Challenge [3] – Track 1 (Classic).

2. Related work

2.1. Video frame interpolation

Deep-learning-based VFI methods can be broadly categorized by their underlying principles. **Flow-based methods** estimate optical flow between input frames and warp pixels to the target timestamp, followed by synthesis and refinement [4, 6, 8, 10, 16, 17]. **Kernel-based methods** predict spatially adaptive convolution kernels to resample input pixels [9]. **Flow-agnostic methods** directly hallucinate intermediate frames without explicit motion estimation, e.g., using 3D convolutions [7]. **Transformer-based methods** leverage attention mechanisms to capture long-range correspondences [21, 26]. EMA-VFI [26] is a representative hybrid approach that uses inter-frame attention to jointly extract motion and appearance features.

From the perspective of motion magnitude, most methods above are designed for and evaluated on moderate-motion benchmarks. To handle large displacements, several approaches have been proposed. ABME [17] strengthens bilateral motion estimation for occlusion-prone and non-linear motion. XVFI [22] adopts a recursive multi-scale shared structure together with complementary flow reversal. FILM [18] employs a multi-scale shared feature extractor and a scale-agnostic bidirectional motion estimation module for large-motion interpolation. SGM [11] adopts sparse global matching to establish long-range correspondences beyond the local receptive field. Despite these advances, extreme-motion scenarios remain challenging, particularly when different spatial regions exhibit diverse motion magnitudes.

2.2. Multi-scale frameworks for VFI

Multi-scale processing is widely used in VFI, primarily as *coarse-to-fine flow estimation*. RIFE [4] introduces IFNet, which progressively refines intermediate flow predictions across multiple resolutions. IFRNet [8] uses a similar pyramid strategy for coarse-to-fine refinement. XVFI [22] and FILM [18] also adopt multi-scale designs, while SAFA [5] explores a related scale-adaptive design in the STVSR setting. Despite their differences, these methods operate on feature-level multi-scale representations within a single forward pass.

Inspired by the effectiveness of these multi-scale designs, we explore a different direction based on image-level multi-scale fusion. Instead of combining intermediate features inside one backbone execution, we run the interpolation backbone at multiple input scales and fuse the resulting full-resolution predictions using learned weights.

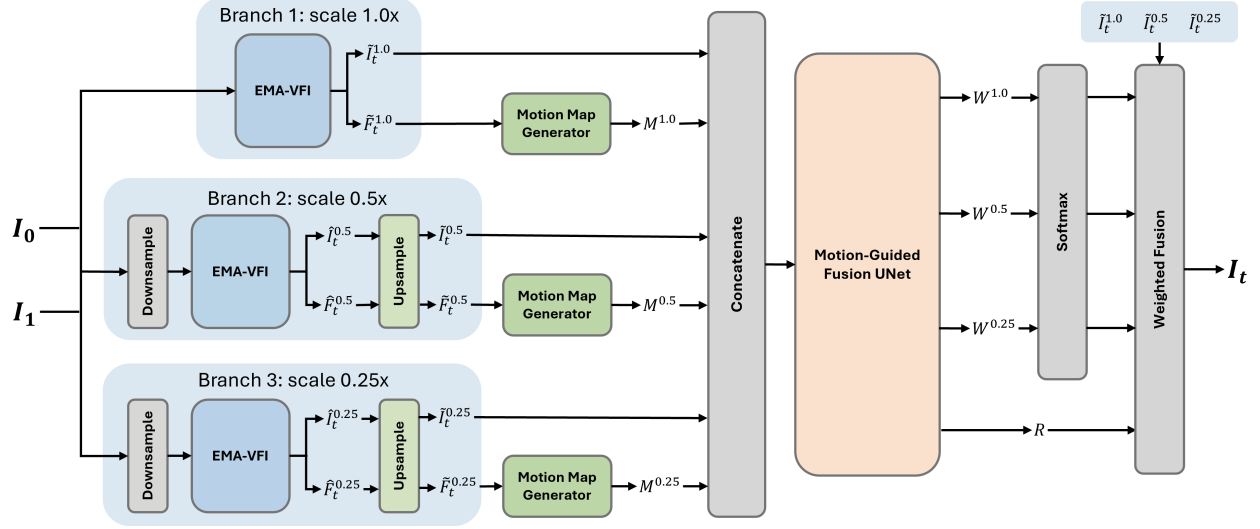


Figure 2. Overview of the proposed motion-guided multi-scale fusion framework. Input frames are processed by EMA-VFI at three scales ($1.0\times$, $0.5\times$, $0.25\times$). Each branch produces an interpolated frame $\hat{I}_t^s$ and bidirectional flow $\hat{F}_t^s$; for downsampled branches these are upsampled to obtain $\tilde{I}_t^s$ and $\tilde{F}_t^s$. Motion maps M^s are derived from the flows. The fusion U-Net takes 15 input channels (9 RGB prediction channels and 6 motion-map channels) and predicts per-scale fusion weights W^s together with a residual R ; softmax normalization yields $\hat{W}^s$, and the final prediction I_t is obtained via weighted fusion with residual correction.

3. Method

3.1. Overview

An overview of the proposed framework is shown in Fig. 2. Given two input frames I_0 and I_1 and a target timestamp $t \in (0, 1)$, our method proceeds in two stages. In the first stage (Sec. 3.2), we perform intermediate frame interpolation at three input scales using EMA-VFI [26] as the backbone, obtaining scale-specific predictions and bidirectional optical flows at the original resolution. In the second stage (Sec. 3.3), a motion-guided fusion module combines the multi-scale predictions into a single output frame by leveraging flow-derived motion maps.

3.2. Multi-scale intermediate frame interpolation

Let $I_0, I_1 \in \mathbb{R}^{3 \times H \times W}$ denote the two input frames. We define a set of scales $\mathcal{S} = \{s_1, s_2, s_3\} = \{1.0, 0.5, 0.25\}$. For each scale $s \in \mathcal{S}$, the processing is as follows.

Downsampling. For $s < 1$, we downsample the inputs using Lanczos interpolation:

$$I_0^s = \mathcal{D}_s(I_0), \quad I_1^s = \mathcal{D}_s(I_1), \quad (1)$$

where $\mathcal{D}_s$ denotes spatial downsampling by factor s , producing frames of size $\lfloor sH \rfloor \times \lfloor sW \rfloor$. For $s = 1.0$, no downsampling is applied.

Backbone interpolation. Each pair (I_0^s, I_1^s) is fed into the EMA-VFI backbone $\mathcal{M}_\theta$, which jointly produces an in-

terpolated frame and a bidirectional optical flow at scale s :

$$(\hat{I}_t^s, \hat{F}_t^s) = \mathcal{M}_\theta(I_0^s, I_1^s, t), \quad (2)$$

where $\hat{I}_t^s$ is the predicted intermediate frame and $\hat{F}_t^s \in \mathbb{R}^{4 \times \lfloor sH \rfloor \times \lfloor sW \rfloor}$ is the concatenated 4-channel bidirectional flow containing forward ($t \rightarrow 0$) and backward ($t \rightarrow 1$) components. We extract the last-level flow from EMA-VFI’s multi-resolution flow list.

Upsampling and flow rescaling. For scales $s < 1$, the predicted frame and flow are upsampled back to the original resolution using bicubic interpolation. Since downsampling the input by factor s reduces pixel displacements proportionally, the flow magnitudes must be rescaled by $1/s$ when mapping back to the original resolution:

$$\tilde{I}_t^s = \mathcal{U}_s(\hat{I}_t^s), \quad (3)$$

$$\tilde{F}_t^s = \frac{1}{s} \mathcal{U}_s(\hat{F}_t^s), \quad (4)$$

where $\mathcal{U}_s$ denotes bicubic upsampling to the original resolution $H \times W$. For $s = 1.0$, we simply set $\tilde{I}_t^s = \hat{I}_t^s$ and $\tilde{F}_t^s = \hat{F}_t^s$. In implementation, the downsampled inputs are padded to multiples of 32 before backbone inference and unpadded before upsampling.

After this stage, we obtain three interpolated frames $\{\tilde{I}_t^s\}_{s \in \mathcal{S}}$ and three bidirectional flow fields $\{\tilde{F}_t^s\}_{s \in \mathcal{S}}$, all at the original resolution.

3.3. Motion-guided multi-scale fusion

The fusion module consists of three components: a motion map generator, a weight and residual estimation network, and a weighted fusion step.

3.3.1. Motion map generator

Different spatial regions may benefit from different interpolation scales: large-motion regions tend to be better handled at coarser scales (where effective displacements are smaller), while fine-texture regions are better preserved at the original scale. To enable the fusion network to make spatially adaptive scale selections, we provide explicit motion-intensity cues derived from the bidirectional optical flow.

For each scale s , the upsampled flow $\tilde{F}_t^s = [\tilde{f}_x^s, \tilde{f}_y^s, \tilde{b}_x^s, \tilde{b}_y^s]$ contains the horizontal and vertical components of the forward and backward flows. We compute the forward and backward motion magnitudes as:

$$m_{\text{fwd}}^s = \log\left(1 + \sqrt{(\tilde{f}_x^s)^2 + (\tilde{f}_y^s)^2 + \epsilon}\right), \quad (5)$$

$$m_{\text{bwd}}^s = \log\left(1 + \sqrt{(\tilde{b}_x^s)^2 + (\tilde{b}_y^s)^2 + \epsilon}\right), \quad (6)$$

where $\epsilon = 10^{-6}$ is a small constant for numerical stability. The logarithmic compression via $\log(1 + \cdot)$ compresses the large dynamic range of motion magnitudes, which can span from near-zero in static regions to hundreds of pixels in extreme-motion areas. The motion map for scale s is then:

$$M^s = [m_{\text{fwd}}^s; m_{\text{bwd}}^s] \in \mathbb{R}^{2 \times H \times W}. \quad (7)$$

3.3.2. Weights and residual estimation

The multi-scale predictions and motion maps are concatenated along the channel dimension to form the fusion input:

$$X = [\tilde{I}_t^{s_1}, \tilde{I}_t^{s_2}, \tilde{I}_t^{s_3}, M^{s_1}, M^{s_2}, M^{s_3}] \in \mathbb{R}^{15 \times H \times W}, \quad (8)$$

where the first nine channels are the three RGB predictions and the last six channels are the three two-channel motion maps.

This tensor is fed into a 5-level U-Net [19] $\mathcal{G}_\phi$ with a base channel width of 32. The encoder consists of five downsampling blocks, each containing two convolutional layers with PReLU activations; the decoder mirrors this structure with transposed convolutions and skip connections. Two separate output heads produce the fusion weight logits and the residual image:

$$(W, R) = \mathcal{G}_\phi(X), \quad (9)$$

where $W \in \mathbb{R}^{3 \times H \times W}$ contains per-scale weight logits and $R \in \mathbb{R}^{3 \times H \times W}$ is the residual correction image.

The per-pixel fusion weights are obtained by applying softmax with a learnable temperature parameter $\tau > 0$:

$$\hat{W}^{s_k} = \frac{\exp(W^{s_k}/\tau)}{\sum_{j=1}^3 \exp(W^{s_j}/\tau)}, \quad k = 1, 2, 3, \quad (10)$$

where τ is parameterized as $\tau = \exp(\lambda)$ with a learnable scalar λ initialized to zero, so that $\tau = 1$ at the start of training and the initial weights are uniform across scales. The exponential reparameterization ensures $\tau > 0$ throughout training.

3.3.3. Weighted fusion

The final interpolated frame is computed as the weighted sum of the multi-scale predictions plus the residual correction:

$$I_t = \text{clamp}\left(\sum_{s \in \mathcal{S}} \hat{W}^s \odot \tilde{I}_t^s + R, 0, 1\right), \quad (11)$$

where $\odot$ denotes element-wise multiplication (broadcast over the RGB channels) and $\text{clamp}(\cdot, 0, 1)$ ensures valid pixel values. The softmax normalization in Eq. (10) guarantees that the weights $\{\hat{W}^s\}_{s \in \mathcal{S}}$ sum to one at every spatial location, so the weighted sum is a convex combination of the scale-specific predictions. The residual R provides an additive correction that can compensate for systematic errors shared across all scales.

4. Experiments

We evaluate the proposed method on multiple benchmarks that cover a range of motion difficulties, and conduct ablation studies to analyze the contribution of each design choice.

4.1. Experimental settings

4.1.1. Datasets

Training data. We use a mixed training set composed of the NTIRE 2026 High-FPS Video Frame Interpolation Challenge data [3] (Tracks 1/2; MIORe/VAR-MIORe [2]) and X4K1000FPS [22]. The combined dataset has 4,934 sequences. We use an approximately 5% sequence-level validation split, yielding 4,688 training and 246 validation sequences. For a fair comparison, all competing methods are initialized from their released checkpoints and finetuned on this same mixed training set while keeping their original architectures unchanged and following their original training setups as closely as possible.

Evaluation benchmarks. We report the main comparison on four challenging benchmarks: SNU-FILM [1] Hard and Extreme, XTest-2K, a derived benchmark obtained by

Table 1. Quantitative comparison after finetuning all methods on the same mixed training data. PSNR $\uparrow$ / SSIM $\uparrow$ / LPIPS $\downarrow$ are reported. Runtime is averaged over SNU-FILM Hard and Extreme at 1280×720 using a single RTX 3090 in FP32. For our method, the reported runtime includes the full three-scale inference and fusion pipeline. Best and second-best results in each column are shown in bold red and underlined blue, respectively.

Method	SNU-Hard			SNU-Extreme			XTest-2K			NTIRE-Val			Params (M) $\downarrow$	Avg. Time (ms) $\downarrow$
	PSNR	SSIM	LPIPS	PSNR	SSIM	LPIPS	PSNR	SSIM	LPIPS	PSNR	SSIM	LPIPS		
XVFI [22]	29.22	0.8868	<u>0.0878</u>	24.83	0.7911	<u>0.1542</u>	29.29	0.8275	0.1297	21.48	0.5826	<u>0.4402</u>	5.61	143.72
FILM [18]	29.27	0.8873	0.0968	24.85	0.7901	0.1595	28.18	0.8143	0.1556	20.94	0.5799	0.4622	34.44	412.06
AMT-G [10]	30.21	0.8984	0.1096	25.56	0.8056	0.1781	31.06	0.8506	<u>0.1106</u>	21.93	0.6080	0.5664	30.64	236.23
EMA-VFI [26]	<u>30.35</u>	<u>0.9024</u>	0.1047	<u>25.68</u>	0.8097	0.1791	<u>31.12</u>	<u>0.8591</u>	0.1130	21.70	<u>0.6135</u>	0.5593	65.66	<u>159.38</u>
SGM-small [11]	29.94	0.8926	0.1123	25.51	0.8023	0.1797	29.91	0.8319	0.1482	<u>22.16</u>	0.6145	0.5353	<u>20.87</u>	308.71
Ours	31.11	0.9087	0.0446	26.06	<u>0.8082</u>	0.0909	32.09	0.8704	0.0451	22.20	0.5840	0.2856	73.89	265.77

2x downsampling XVFI’s X-TEST set [22], and NTIRE-Val, which is the validation portion of the NTIRE 2026 challenge Track 1 and Track 2 training data containing extreme-motion sequences. These benchmarks focus on difficult motion regimes and provide a targeted assessment of interpolation quality under challenging conditions.

Metrics. We measure PSNR and SSIM [24] for distortion-oriented evaluation, and LPIPS [27] (AlexNet backbone) for perceptual quality assessment.

4.1.2. Implementation details

Model configuration. The base interpolator is EMA-VFI [26] with the default architecture ($F=32$, $W=7$, $\text{depth}=[2, 2, 2, 4, 4]$), containing 65.66M parameters. The fusion U-Net has a base channel width of 32 and takes 15 input channels in total, consisting of 9 RGB prediction channels and 6 motion-map channels. It predicts 3-channel weight logits and a 3-channel residual image, adding approximately 8.23M parameters. The total model size is 73.89M parameters. Input downsampling uses Lanczos interpolation; upsampling uses bicubic interpolation.

Training procedure. Training is conducted in two stages:

1. **Stage 1** (200 epochs): The EMA-VFI backbone is frozen and initialized from official pre-trained weights. Only the fusion U-Net is trained.
2. **Stage 2** (200 epochs): The entire model is jointly fine-tuned with differential learning rates— 10^{-5} for the EMA-VFI backbone and 10^{-4} for the fusion module.

We use AdamW [13] with weight decay 10^{-4} and a cosine learning-rate schedule [12] (initial learning rate 2×10^{-4} for Stage 1). The training crop size is 512×512 with a batch size of 2. Gradient clipping with a maximum norm of 1.0 is applied. Training is performed in FP32 on NVIDIA RTX 3060 GPUs.

Data augmentation. We apply random cropping, random horizontal and vertical flips, random channel permutation, random temporal reversal, and random rotation.

Loss function. We employ a mixed loss with dynamic normalization across terms:

$$\mathcal{L} = w_{\text{lap}} \mathcal{L}_{\text{lap}} + w_{\text{lpips}} \mathcal{L}_{\text{lpips}} + w_{\text{ms-ssim}} (1 - \mathcal{L}_{\text{MS-SSIM}}), \quad (12)$$

where $\mathcal{L}_{\text{lap}}$ is a 5-level Laplacian pyramid L1 loss [4, 16], $\mathcal{L}_{\text{lpips}}$ is the LPIPS perceptual loss [27] with an AlexNet backbone, and $\mathcal{L}_{\text{MS-SSIM}}$ is the multi-scale structural similarity [23]. In our experiments, we set $w_{\text{lap}} = 1.0$, $w_{\text{lpips}} = 0.5$, and $w_{\text{ms-ssim}} = 0.5$. Each loss term is dynamically rescaled so that its gradient contribution is proportional to the Laplacian loss magnitude, preventing any single term from dominating.

4.2. Comparison with representative methods

We compare the proposed method with several representative VFI approaches: XVFI [22], FILM [18], AMT-G [10], EMA-VFI [26], and SGM-small [11]. For a fair comparison, all methods are initialized from their released checkpoints and finetuned on the same mixed training set described above.

Quantitative results. Tab. 1 reports PSNR, SSIM, and LPIPS on the four target benchmarks together with parameter count and average runtime. For runtime, we report the average over SNU-FILM Hard and Extreme at 1280×720 . After finetuning all methods on the same mixed training data, our method achieves the best PSNR and LPIPS on all four benchmarks, and also obtains the best SSIM on SNU-Hard and XTest-2K. EMA-VFI and SGM-small obtain higher SSIM on SNU-Extreme and NTIRE-Val, respectively, but our method still provides clearly better PSNR–LPIPS trade-offs on both benchmarks.

The multi-scale design introduces additional computational overhead because the backbone is executed at three input scales. Our model contains 73.89M parameters and runs in 265.77 ms on average under the SNU-FILM setting. It is slower than XVFI, EMA-VFI, and AMT-G, but remains faster than FILM and SGM-small while delivering the best overall reconstruction quality on the reported benchmarks.

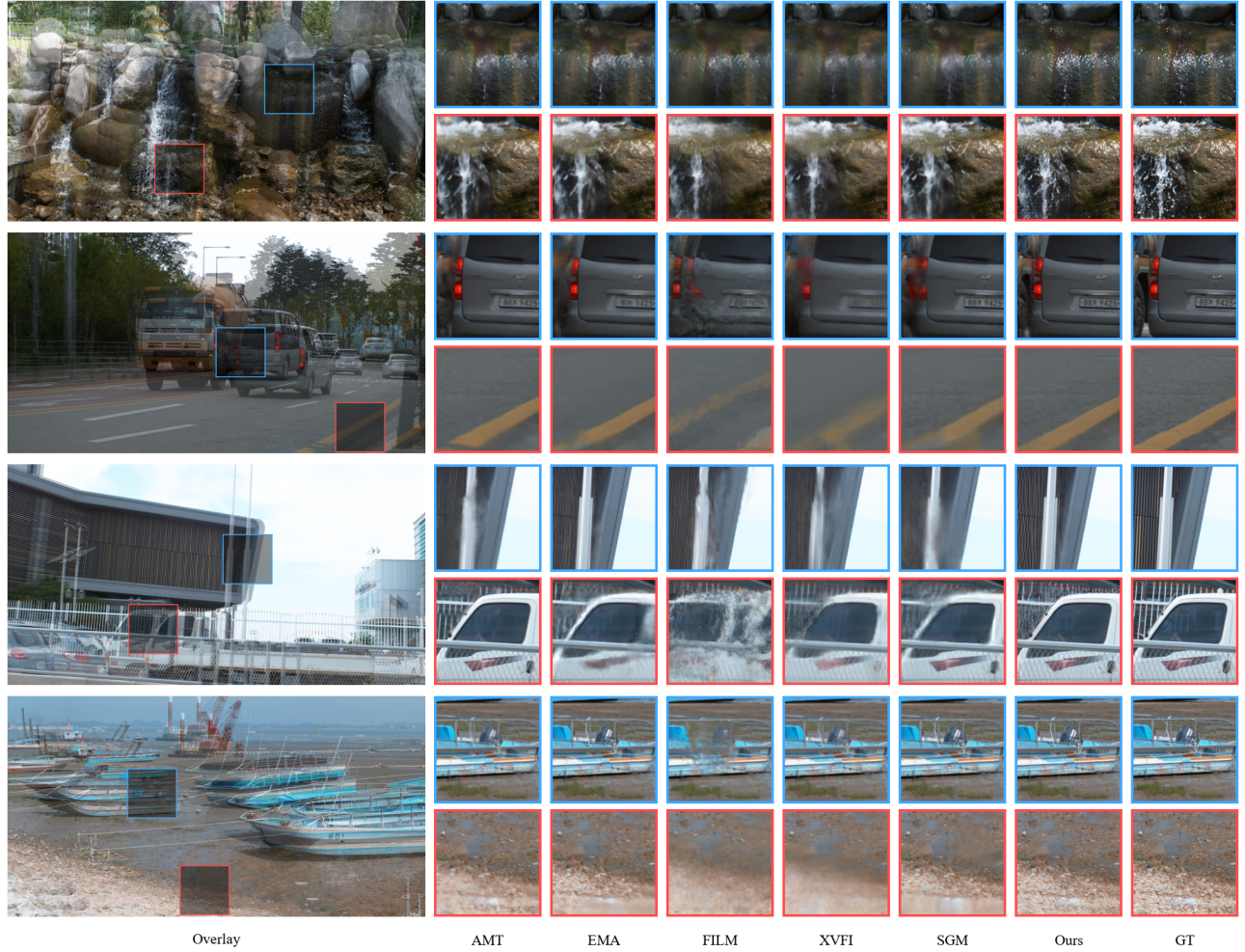


Figure 3. Qualitative comparison on large-motion scenes from XTest-2K.

Qualitative results. Fig. 3 presents qualitative comparisons on large-motion scenes from XTest-2K. Compared with the competing methods, our approach reconstructs clearer object boundaries in large-displacement regions while preserving sharper local details in relatively static areas. The visual differences are most evident near motion boundaries, where adaptive fusion can place higher emphasis on coarser branches while retaining finer-scale details in nearby static content.

4.3. Ablation study

We conduct ablation experiments on XTest-2K and NTIRE-Val to examine the contributions of scale settings, motion-map guidance, and loss design, and further analyze backbone dependency and computational overhead. These two benchmarks provide complementary views: XTest-2K reflects generalization to a challenging set, while NTIRE-Val reflects behavior on the challenge validation data.

Table 2. Ablation on scale settings.

Method	XTest-2K			NTIRE-Val		
	PSNR	SSIM	LPIPS	PSNR	SSIM	LPIPS
1-Scale	31.34	0.8557	0.0513	21.36	0.5683	0.3023
2-Scale	32.05	0.8704	0.0429	21.98	0.5832	0.2809
3-Scale (Ours)	32.09	0.8704	0.0451	22.20	0.5840	0.2856

4.3.1. Comparison of scale configurations

Tab. 2 compares single-scale ($1.0\times$), two-scale ($1.0\times$, $0.5\times$), and three-scale ($1.0\times$, $0.5\times$, $0.25\times$) configurations. The 1-Scale baseline directly applies EMA-VFI at the original resolution without the fusion U-Net, while the 2-Scale and 3-Scale variants use the same fusion strategy and differ only in the number of input scales. All these variants are trained with the loss used in this paper rather than the original EMA-VFI loss.

Adding coarser branches consistently improves PSNR

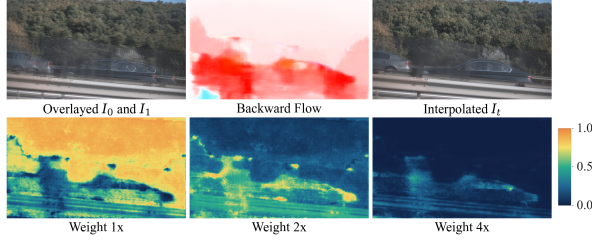


Figure 4. Visualization of learned fusion weights for different input scales. Coarser branches receive higher weights in large-motion regions, while the $1\times$ branch remains more active in relatively static areas.

Table 3. Ablation on motion-map guidance.

Method	XTest-2K			NTIRE-Val		
	PSNR	SSIM	LPIPS	PSNR	SSIM	LPIPS
Without Motion Map	32.03	0.8689	0.0464	22.12	0.5823	0.2760
With Motion Map	32.09	0.8704	0.0451	22.20	0.5840	0.2856

over the single-scale baseline. The 3-Scale setting yields the highest PSNR on both benchmarks and the highest SSIM on NTIRE-Val, while the 2-Scale setting achieves the lowest LPIPS on both benchmarks. This suggests that additional coarse branches strengthen distortion-oriented accuracy, whereas the best perceptual operating point may vary slightly with the scale set. As shown in Fig. 4, the learned fusion weights emphasize coarser branches around large-motion regions while preserving stronger $1\times$ responses in relatively static areas, providing visual evidence for the effectiveness of motion-guided multi-scale fusion.

4.3.2. Effect of motion-map guidance

Tab. 3 compares the full model with a variant that removes the six motion-map channels and feeds only the three scale-specific predictions to the fusion module. Motion guidance improves PSNR and SSIM on both benchmarks and slightly lowers LPIPS on XTest-2K, while the LPIPS change on NTIRE-Val is mixed. This suggests that motion maps mainly help the fusion module pick more reliable scales in motion-varying regions.

4.3.3. Effect of loss design

Tab. 4 examines the impact of different loss configurations. Starting from the fully trained 3-Scale model, we continue training for 200 epochs with MSE-only and MSE+MS-SSIM losses to explore alternative loss formulations aimed at maximizing PSNR.

The MSE and MSE+MS-SSIM variants improve NTIRE-Val PSNR to 22.71 and also raise SSIM, but at the cost of substantially degraded LPIPS. The full loss (Laplacian + LPIPS + MS-SSIM) yields markedly better LPIPS on both benchmarks, indicating a better distortion-perception balance in our setting. This indicates that the mixed loss

Table 4. Ablation on loss design. All variants use the 3-Scale architecture.

Loss	XTest-2K			NTIRE-Val		
	PSNR	SSIM	LPIPS	PSNR	SSIM	LPIPS
MSE	32.06	0.8727	0.0948	22.71	0.6182	0.5320
MSE+MS-SSIM	31.94	0.8758	0.0993	22.71	0.6205	0.5332
Full Loss (Ours)	32.09	0.8704	0.0451	22.20	0.5840	0.2856

Table 5. Applying the proposed fusion design to AMT.

Method	XTest-2K			NTIRE-Val		
	PSNR	SSIM	LPIPS	PSNR	SSIM	LPIPS
AMT	31.06	0.8506	0.1106	21.93	0.6080	0.5664
AMT + Fusion	31.16	0.8423	0.1580	22.31	0.6098	0.5315

is more suitable for the perceptual objective emphasized in our setting.

4.4. Applying the fusion design to AMT

To further analyze backbone dependency, we apply the same motion-guided multi-scale fusion design to AMT [10] and denote the resulting variant as AMT + Fusion. As shown in Tab. 5, AMT + Fusion improves PSNR on both XTest-2K and NTIRE-Val, and also improves SSIM and LPIPS on NTIRE-Val, but yields weaker SSIM and LPIPS on XTest-2K. This result suggests that the benefit of the proposed multi-scale fusion design is not limited to EMA-VFI, although the final distortion-perception trade-off remains dependent on the underlying backbone.

4.5. Limitations and discussion

The main limitation of the proposed framework is the additional inference cost introduced by repeated multi-scale backbone execution. Although the fusion U-Net itself is relatively lightweight, the overall runtime still increases because the EMA-VFI backbone is executed at three input scales.

We reuse the same backbone across all scales instead of introducing separate scale-specific models, which keeps the parameter overhead moderate (73.89M vs. 65.66M for EMA-VFI). This design simplifies training and implementation, but does not remove the cost of repeated multi-scale inference.

The AMT + Fusion results further suggest that the proposed fusion design can also improve distortion-oriented accuracy on another strong interpolation backbone, while the final distortion-perception trade-off remains backbone dependent. Future improvements may come from sharing features across scales, dynamically selecting active scales, or distilling the multi-scale model into a faster single-pass model.

5. Conclusion

We presented a motion-guided multi-scale fusion framework for High-FPS video frame interpolation in a large-motion regime. By running EMA-VFI at three resolutions and fusing the resulting predictions with a lightweight U-Net guided by flow-derived motion maps, the method performs spatially adaptive scale selection that better matches local motion complexity across image regions. Experiments on challenging benchmarks show clear gains over strong baselines, especially in the overall PSNR–LPIPS trade-off in scenes containing regions with different motion magnitudes. Applying the same fusion design to AMT further suggests that the method is not tied to a single backbone, although the final trade-off remains backbone dependent. The main limitation is the additional inference cost of repeated multi-scale backbone execution at test time. Future work will explore feature sharing across scales, dynamic routing, and distillation to faster single-pass models.

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